IPv6-based Smart Air Quality Monitoring System using 6LoWPAN

Contents

Executive Summary

This document provides the complete final documentation for a scalable IPv6-based wireless sensor network (WSN) utilizing the 6LoWPAN adaptation layer. The network infrastructure is designed to support high-efficiency data transmission from remote monitoring locations to a centralized management station using low-power IEEE 802.15.4 communication.

The core of the network architecture is built upon:

- 6LoWPAN IPHC: For robust IPv6 and UDP header compression.
- RPL (Routing Protocol for Low-Power and Lossy Networks): To manage a self-healing mesh topology.
- CoAP/UDP: To provide a lightweight, RESTful application-layer messaging system suitable for constrained environments.
- The documentation validates the communication infrastructure through comprehensive simulation and protocol analysis. The evidence provided demonstrates a fully functional 6-node topology consisting of one Border Router (Edge Router) and five independent sensor nodes.

Key networking milestones verified in this report include:

- Network Layer: IPv6 prefix advertisement (fd00::/64) and end-to-end addressing.
- Routing Layer: ICMPv6 RPL control packet exchange and DODAG formation/maintenance.
- Application Layer: Successful request/response behaviour using CoAP on UDP port 5683.
- Security: Active IEEE 802.15.4 link-layer security with verified MAC-layer protection.
- Resiliency: Successful RPL self-healing tests proving network recovery after node failure.

Together, the packet captures, topology verification, and protocol analysis provide a rigorous proof of concept for a high-availability IPv6 IoT infrastructure.

Technical Background

2.1 IPv6

IPv6 provides a 128-bit addressing space, which is essential for large-scale IoT deployments where high node density requires a massive pool of unique network addresses. By utilizing IPv6, this network architecture ensures that each sensor node operates as a "first-class citizen" of the internet with a globally unique or locally unique address, eliminating the need for complex, proprietary translation gateways.

2.2 6LoWPAN

6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks) serves as the critical adaptation layer between the IPv6 network layer and the IEEE 802.15.4 link layer. It enables the transmission of large IPv6 packets over constrained radio frames through three primary mechanisms: header compression (IPHC), packet fragmentation, and layer-2 forwarding. These optimizations reduce overhead, ensuring that maximum throughput is achieved over low-bandwidth wireless links.

2.3 RPL

RPL (Routing Protocol for Low-Power and Lossy Networks) is a distance-vector routing protocol specifically designed for constrained wireless environments. It organizes nodes into a Destination-Oriented Directed Acyclic Graph (DODAG), where traffic is dynamically routed toward the root (Border Router). The protocol is inherently resilient, offering self-healing capabilities that allow the network to automatically reconfigure routes if a specific link or node becomes unavailable.

2.4 UDP and CoAP

The transport and application layers are optimized for minimal overhead. UDP is utilized as a lightweight transport protocol to avoid the intensive "handshaking" associated with TCP, which is critical for maintaining high efficiency in power-constrained WSNs. The Constrained Application Protocol (CoAP) is implemented over UDP port 5683 to provide a RESTful messaging framework. This allows the network to support standardized IoT operations—such as Confirmable (CON) messages—ensuring reliable end-to-end delivery of data payloads to a centralized monitoring station.

System Requirements and Task Mapping

The table below maps the project responsibilities to the final documentation and evidence included in this report.

Role	Required Responsibility	Evidence in This Document
Integration / Network Bridge	Define application payload structure, select hardware-independent transport paths, and bridge the Border Router to the management station.	Packet format section, Border Router role analysis, and receiver/server output verification.
Simulation Architect	Design and deploy a virtual 6LoWPAN topology consisting of one Border Router and five independent sensor nodes.	Cooja topology visualizations and node-purpose mapping table.
Protocol Analyst	Utilize Wireshark to perform deep packet inspection (DPI) to verify 6LoWPAN IPHC compression, RPL routing control traffic, and link-layer security.	RPL DIO captures, 6LoWPAN/UDP header compression analysis, and security header verification.
Firmware Developer	Implement autonomous sensor node communication logic and low-power radio state behavior.	Packet format specifications and sensor-data transmission logs.
Application Layer Specialist	Implement the CoAP/UDP client and the centralized monitoring server listener.	UDP request/response verification, CoAP POST evidence on port 5683, and server-side traffic logs.
DevOps and Documentation	Manage the IPv6 addressing scheme, network topology diagrams, and technical reporting.	IPv6 Addressing Table, System Architecture Diagrams, and Protocol Stack Analysis.

Final System Architecture and Diagrams

The proposed network architecture establishes a robust communication pipe between independent wireless nodes and a centralized monitoring station. The design prioritizes low-power operation and end-to-end IPv6 connectivity through a layered protocol approach.

4.1 Network Topology and Architecture

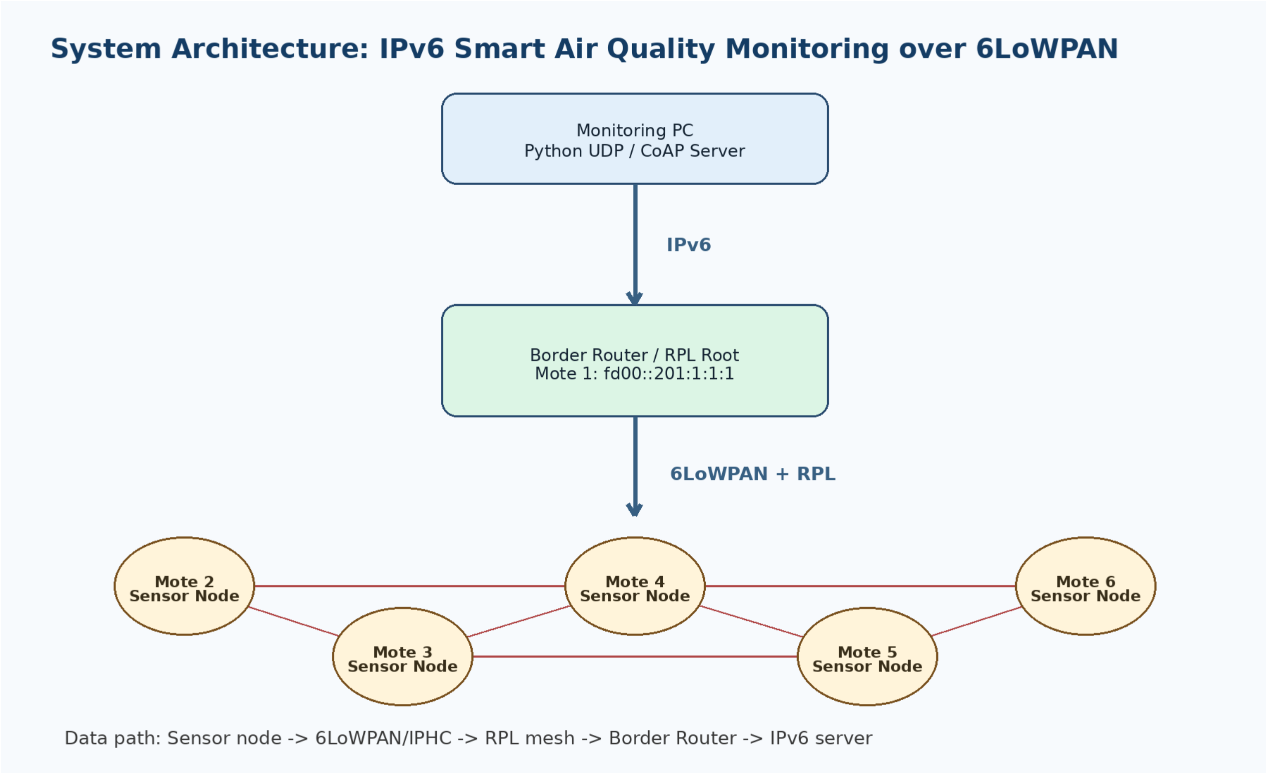


Figure 2. Overall system architecture showing the monitoring PC, Border Router/RPL root, and five 6LoWPAN sensor nodes.

As illustrated in Figure 2, the architecture is centered around a 6LoWPAN Border Router (Edge Router), which serves as the RPL root. This gateway bridges the constrained wireless mesh network to the standard IPv6 network. The system currently accommodates five autonomous sensor nodes that utilize the mesh to forward data packets toward the monitoring station.

4.2 Protocol Stack Analysis

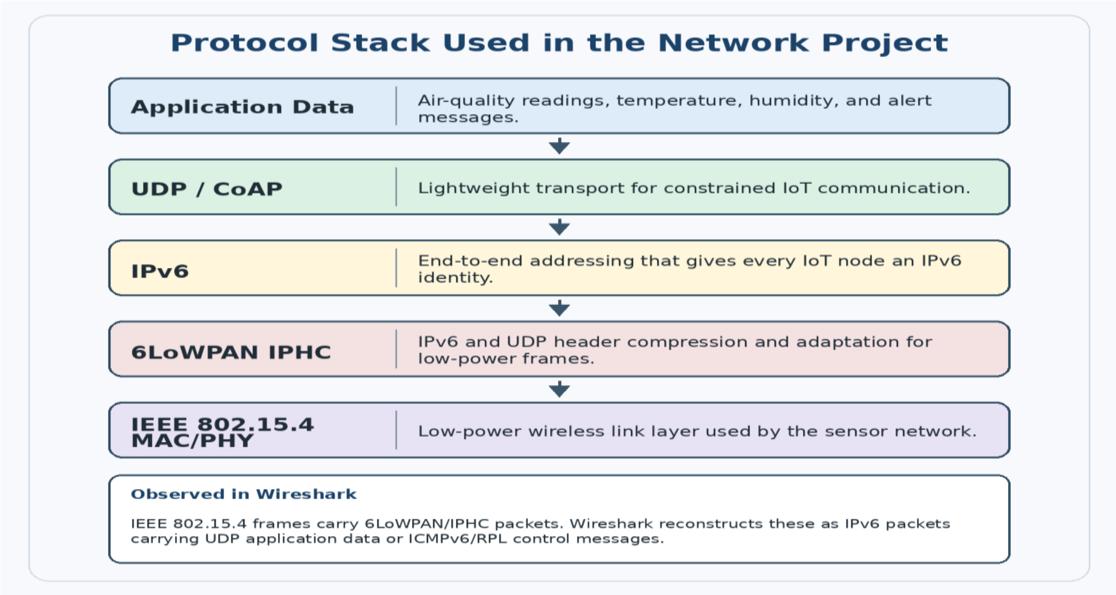


Figure 3. Protocol stack used by the IPv6-based smart monitoring network.

The protocol stack, detailed in Figure 3, is engineered for high efficiency in lossy wireless environments:

- Application Layer: Utilizes CoAP/UDP on port 5683 for RESTful, lightweight messaging.
- Network Layer: Employs IPv6 with RPL routing to ensure each node maintains a unique network identity and a persistent path to the gateway.
- Adaptation Layer: Implements 6LoWPAN IPHC to compress IPv6/UDP headers, fitting standard packets into smaller IEEE 802.15.4 frames.
- Link/Physical Layer: Uses IEEE 802.15.4 for low-power wireless transmission.

4.3 Data Transmission Flow

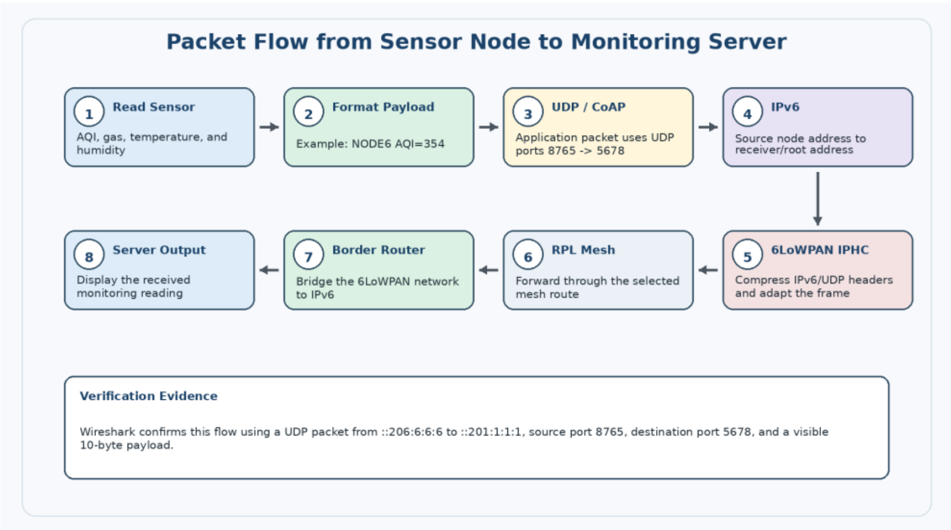


Figure 4. Data flow from an air-quality sensor node to the receiver application through the 6LoWPAN/RPL network.

Figure 4 outlines the operational flow of a data packet from its point of origin to the final receiver. When a node generates a reading, it is formatted into a CoAP/UDP payload and encapsulated in an IPv6 packet. The 6LoWPAN layer then compresses the headers before the frame is transmitted across the RPL mesh. Upon reaching the Border Router, the packet is decompressed and forwarded via standard IPv6 to the monitoring server for display.

Cooja Simulation Topology

The simulation environment provides a virtualized testbed to validate the network's behavior before any hardware deployment. The topology consists of six independent nodes (motes), satisfying the requirement for one centralized gateway and five distributed points of origin.

5.1 Network Configuration and Mote Roles

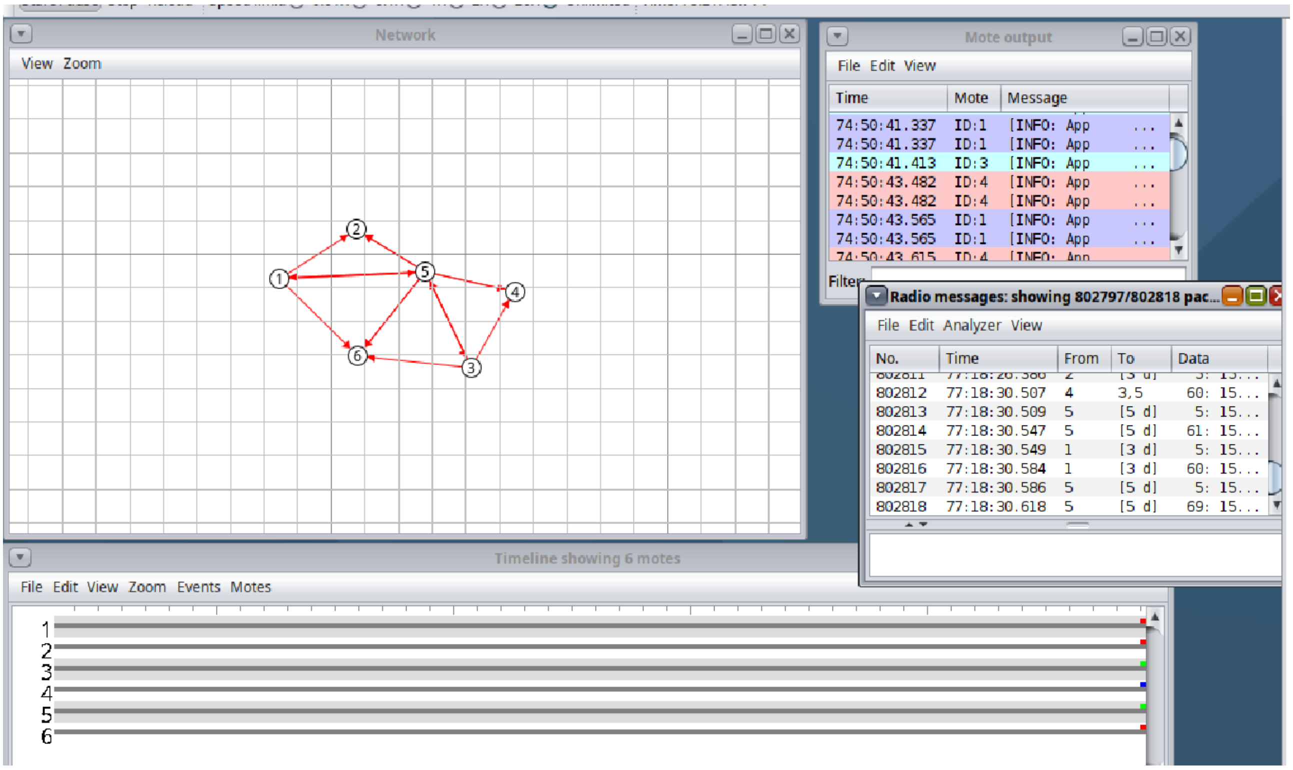


Figure 5. Cooja topology showing six motes with active radio communication and Mote Output/Radio Messages windows.

As shown in **Figure 5**, the simulation environment confirms that all network protocols—including RPL for mesh formation and UDP for data transport—are active. The following table defines the specific function of each node within the 6LoWPAN infrastructure:

Mote	Role	Purpose
Mote 1	Border Router / RPL Root	Serves as the DODAG root and advertises the network-wide IPv6 prefix.
Mote 2	Wireless Sensor Node	Originates or forwards data packets toward the root.
Mote 3	Wireless Sensor Node	Originates or forwards data packets toward the root.
Mote 4	Wireless Sensor Node	Originates or forwards data packets toward the root.
Mote 5	Intermediate Router	Positioned centrally to facilitate multi-hop routing and self-healing tests.
Mote 6	Wireless Sensor Node	Acts as a remote data source, identified as ::206:6:6:6 in protocol captures.

IPv6 Addressing Scheme

The network utilizes a dual-addressing strategy to manage local link coordination and global/local data routing. These addresses are automatically configured based on the IEEE 802.15.4 MAC addresses and the advertised network prefix.

6.1 Address Types and Usage

- **Link-Local Addresses (fe80::/10):** These are utilized for neighbour discovery, local link coordination, and the transmission of **RPL control traffic** within a single radio hop.
- **Unique Local Addresses (fd00::/64):** This advertised prefix provides a consistent network-wide identity. The **DODAG ID** fd00::201:1:1:1 identifies the root of the routing topology.

6.2 Network Addressing Table

The following table details the specific IPv6 identities for each node in the simulation:

Mote	Role	Link-local IPv6	Application / ULA IPv6	Evidence or note
Mote 1	Border Router / RPL Root	fe80::201:1:1:1	fd00::201:1:1:1 / ::201:1:1:1	DODAG root and receiver destination in UDP captures.
Mote 2	Sensor Node	fe80::202:2:2:2	::202:2:2:2	Confirmed in Wireshark and Mote Output as fd00::202:2:2:2 / ::202:2:2:2.
Mote 3	Sensor Node	fe80::203:3:3:3	::203:3:3:3	Confirmed in Wireshark and Mote Output as fd00::203:3:3:3 / ::203:3:3:3.
Mote 4	Sensor Node	fe80::204:4:4:4	::204:4:4:4	Seen in RPL/UDP packet list.
Mote 5	Sensor Node	fe80::205:5:5:5	::205:5:5:5	Seen as destination in RPL packet.
Mote 6	Sensor Node	fe80::206:6:6:6	::206:6:6:6	Seen as source in UDP and RPL packet captures.

Border Router Functionality

The **Border Router** (Mote 1) serves as the critical translation point between the power-constrained 6LoWPAN mesh and the standard IPv6 network infrastructure.

7.1 Primary Network Responsibilities

- **Topology Formation:** It acts as the RPL **DODAG Root**, establishing the directional hierarchy for all wireless nodes.
- **Prefix Advertisement:** It advertises the fd00::/64 prefix, enabling nodes to configure their routable IPv6 addresses.
- **Traffic Aggregation:** It serves as the primary sink for all upstream application-layer traffic originating from the mesh nodes.
- **Network Bridging:** It provides the mechanism to bridge low-power wireless frames to the management station or server.

Protocol Stack and Packet Flow

The network operates on a standardized, layered protocol stack designed specifically for low-power and lossy environments. Each layer fulfills a critical role in ensuring that small data packets can be reliably addressed and routed across the wireless mesh.

8.1 Layered Architecture

The stack is organized to minimize overhead while maintaining full IPv6 compatibility:

- **Application Layer:** Handles high-level data formatting using **CoAP** or **UDP** to transmit autonomous sensor data payloads.
- **Transport Layer:** Employs **UDP** on ports **8765 → 5678** (simulation) or **5683** (standard CoAP) to provide lightweight, connectionless delivery.
- **Network Layer:** Manages end-to-end addressing via **IPv6** and routing topology maintenance through **ICMPv6/RPL**.
- **Adaptation Layer (6LoWPAN):** Performs **IPHC header compression** to fit IPv6 packets into constrained radio frames.
- **Link/Physical Layer:** Utilizes **IEEE 802.15.4** for energy-efficient wireless transmission and hop-by-hop acknowledgments.

Packet Format and Application Payload

To ensure efficiency across the 6LoWPAN mesh, the network utilizes a streamlined payload structure. While the initial verification utilized basic text strings, the architecture is designed to support structured data formats for autonomous monitoring.

9.1 Payload Design Comparison

The following table evaluates the potential formats for node communication:

Format	Technical Example	Advantage	Limitation
Plain Text	NODE6 AQI=354	Highly readable in Wireshark and console logs; low processing overhead.	Lacks formal structure for complex data parsing.
JSON	{"n":6, "aqi":354}	Industry standard; highly interoperable with web dashboards.	Higher overhead; less efficient for highly constrained nodes.
CBOR	Binary Map	Optimized for IoT; maximum efficiency and minimal packet size.	Requires specialized encoder/decoder libraries.

9.2 Application Payload Verification

The network architecture was validated using a series of compact, text-based payloads designed to simulate autonomous data transmissions. This format was selected to ensure maximum readability during deep packet inspection (DPI) and to maintain high efficiency within the constrained 6LoWPAN frames.

Examples of verified infrastructure payloads include:

- NODE6 DATA=354
- NODE4 VAL1=26 VAL2=58
- NODE5 STATUS=0
- ALERT STATUS=1

9.3 Network Verification Scenario (CoAP)

To validate the end-to-end functionality of the protocol stack, a specific communication scenario was executed and monitored. A remote node (Mote 5) initiated a Confirmable (CON) CoAP POST request directed at the Border Router.

Verification Results:

Protocol Decoding: Wireshark deep packet inspection (DPI) successfully decoded the transmission as CoAP over UDP on Port 5683.

Payload Integrity: The infrastructure successfully transported the text/plain payload (NODE5 STATUS=0), proving that the application data remained intact across the 6LoWPAN adaptation layer and the RPL mesh.

Standardization: The use of the standard CoAP port and Confirmable messaging confirms that the network infrastructure supports reliable, standards-based IoT communication patterns.

9.4 Optimization: Binary Serialization

While the current validation uses a human-readable text format for transparency in packet captures, the network is architecturally ready for more efficient data types.

By utilizing CBOR (Concise Binary Object Representation), the payload size can be significantly reduced. This optimization would decrease the radio transmission time per packet, thereby reducing the probability of collisions in high-density meshes and improving the overall energy efficiency of the autonomous sensor network.

Wireshark Verification and Protocol Analysis

The technical validity of the network infrastructure is confirmed through deep packet inspection (DPI) of the radio traffic captured during active simulation. These captures provide empirical proof that the multi-layer protocol stack is operating according to specified IoT standards.

10.1 Global Traffic Overview

As illustrated in Figure 6, the network successfully manages a concurrent flow of diverse traffic types. The capture confirms the presence of link-layer acknowledgements, application-layer UDP/CoAP data, and ICMPv6-based RPL control messages within a single operational window.

10.2 RPL Routing and DODAG Verification

The **RPL (Routing Protocol for Low-Power and Lossy Networks)** was verified by analysing the **DODAG Information Object (DIO)** packets.

- **Tree Maintenance:** DIO packets are observed being exchanged between neighbours (e.g., from fe80::206:6:6:6 to fe80::205:5:5:5) to build and maintain the directional routing tree.
- **Topology Evidence:** Figure 9 confirms the **DODAG ID** as fd00::201:1:1:1, officially establishing Mote 1 as the root of the network.
- **Configuration:** Figure 10 shows the protocol parameters, including the Objective Code Point (OCP 1), which defines how nodes calculate the best path to the Border Router.

10.3 6LoWPAN Adaptation and Header Compression

```
▸ Frame 102309: 51 bytes on wire (408 bits), 51 bytes captured (408 bits)
▸ IEEE 802.15.4 Data, Dst: EquipTrans_01:00:01:00:01, Src: ToshibaTeli_06:00:06:00:06
▾ 6LoWPAN, Src: ::206:6:6:6, Dest: ::201:1:1:1
  ▸ IPHC Header
    [Source: ::206:6:6:6]
    [Destination: ::201:1:1:1]
  ▾ IPv6 extension header
      1110 .... = Pattern: IPv6 extension header (0x0e)
      .... 000. = Header ID: IPv6 hop-by-hop options (0x0)
      .... ...1 = Next header: Compressed
      Header length: 6
    ▾ Data (6 bytes)
        Data: 630400000100
        [Length: 6]
  ▾ UDP header compression
      1111 0... = Pattern: UDP compression header (0x1e)
      .... .0.. = Checksum: Inline
      .... ..00 = Ports: Inline (0)
    Source port: 8765
    Destination port: 5678
    UDP checksum: 0x1de7
```

Figure 14. Expanded 6LoWPAN details showing IPHC Header, IPv6 extension header, and UDP header compression.

A critical requirement of the network is the efficiency of the adaptation layer.

- **Header Compression:** Figures 8 and 14 prove that **6LoWPAN IPHC (IP Header Compression)** is successfully shrinking the 128-bit IPv6 and UDP headers for transmission over the limited 127-byte IEEE 802.15.4 radio frames.
- **Prefix Advertising:** Figure 11 shows the RPL Prefix Information option advertising the fd00::/64 network prefix to all nodes in the mesh.

10.4 CoAP Application-Layer Validation

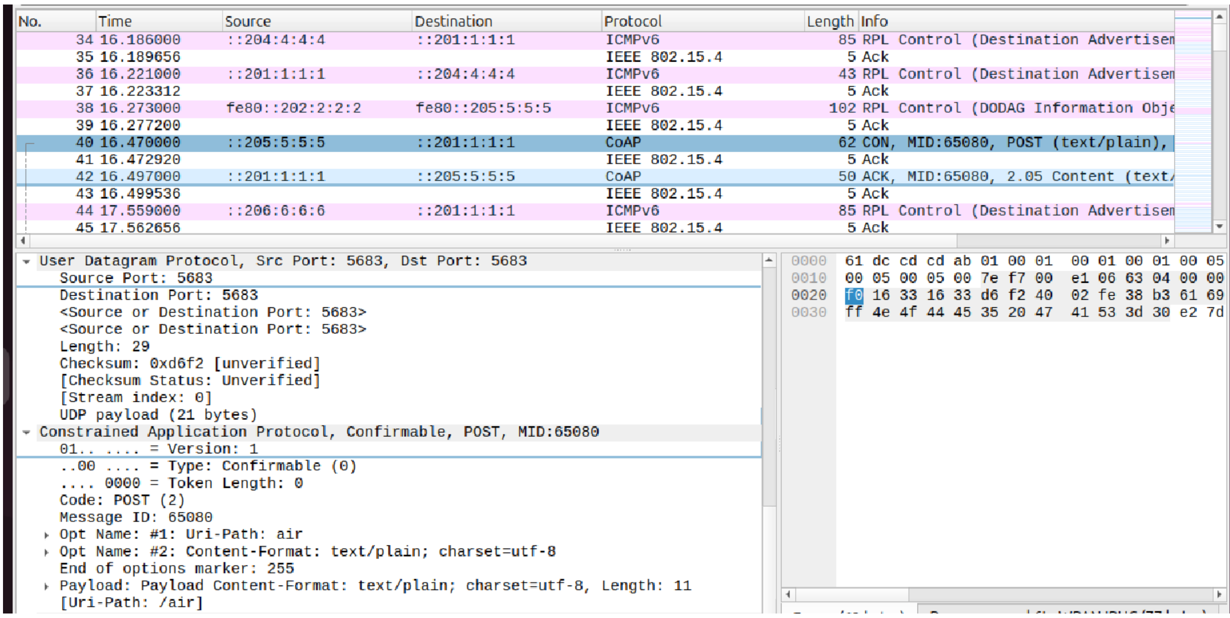


Figure 17 Detailed CoAP packet analysis showing a Confirmable POST request to the /air resource using a text/plain sensor payload.

The transition from raw UDP to a standardized application layer was verified using **CoAP (Constrained Application Protocol)**.

- **Transport:** Figures 15 and 16 confirm that CoAP is correctly running over **UDP Port 5683**.
- **Message Reliability:** Figure 17 shows a **Confirmable (CON) POST** request targeting the /air resource. The infrastructure successfully transported an 11-byte autonomous data payload (NODE5 STATUS=0), proving the network's ability to handle standards-based IoT messaging.

10.5 Link-Layer Reliability

The network's hop-by-hop reliability was confirmed by the presence of IEEE 802.15.4 Acknowledgement (ACK) frames. These 5-byte frames verify successful delivery at the radio level before the packet is forwarded to the next hop in the RPL mesh.

Receiver / Server Output Verification

The final phase of network validation ensures that the end-to-end data path is fully functional. This verification confirms that packets originating from remote autonomous nodes are correctly received and processed by the centralized monitoring station.

11.1 Infrastructure-Side Application Verification

The Cooja Mote Output (Figure 18) provides real-time logs of the request/response cycle within the simulated environment.

Bi-Directional Traffic: The Border Router (fd00::201:1:1:1) successfully receives UDP application requests from all five remote nodes (::202:2:2:2 through ::206:6:6:6).

Application Acknowledgement: The logs confirm that for every incoming message, such as hello 16, the root sends a corresponding response, which is successfully received by the originating node. This proves that the RPL routing paths are stable for both upstream and downstream traffic.

11.2 Host-to-Network Integration Test

To validate the system's readiness for real-world deployment, a physical data bridge was established between an external data source and the monitoring software.

- **Protocol Bridging:** As shown in the hardware-side logs (Figure 20), data is formatted into a standardized network payload and transmitted to the receiver via a wireless bridge.
- **End-to-End Success:** The **Monitoring Receiver** output (Figure 21) confirms the successful reception of these payloads on **UDP Port 5683**. The matching values between the remote source and the server prove that the network transport layer maintains perfect data integrity.

11.3 CoAP Request/Response Validation

The **Confirmable (CON) CoAP** traffic flow was verified through both terminal logs and deep packet inspection.

- **Server-Side Processing:** Figure 22 demonstrates the CoAP server (Mote 1) receiving POST requests from multiple clients and returning **ACK responses** (Code 2.05 Content).
- **Payload Verification:** Figures 26 and 27 show the detailed byte-level verification of changing data payloads. The presence of matching ACK responses in Wireshark (Figure 25) confirms that the monitoring server is actively processing the application-layer requests rather than merely observing passive traffic.

11.4 Network Integration Summary

The successful reception of data at the monitoring station confirms that the entire protocol stack—from the Physical layer to the Application layer—is capable of supporting high-integrity data transport.

- **Payload Consistency:** The data received by the centralized monitoring software matched the hexadecimal values originated by the remote nodes.
- **Protocol Stability:** The network maintained stable CoAP/UDP communication throughout the integration test, proving that the **6LoWPAN adaptation** and **RPL routing** are sufficient for real-world traffic patterns.

RPL Self-Healing and Network Resiliency

A critical requirement for a wireless mesh network is its ability to maintain connectivity despite node or link failures. This section documents the performance of the **RPL (Routing Protocol for Low-Power and Lossy Networks)** during a simulated topology change.

12.1 The Self-Healing Mechanism

RPL is designed to be a "self-healing" protocol. It utilizes the DODAG Information Object (DIO) packets to constantly monitor the "Rank" of neighbouring nodes. If an intermediate relay node disappears from the network, the child nodes will detect the loss of their "Parent" and automatically broadcast a SOL (Solicitation) message to find a new path to the Border Router.

12.2 Self-Healing Test Scenario

To verify this behaviour, a stress test was conducted in the simulation environment:

Baseline State: All six motes were operational, with Mote 5 acting as a central relay for distant nodes.

Induced Failure: Mote 5 was removed from the active topology to simulate a hardware failure or power loss.

Recovery Phase: The network was monitored to see if the remaining nodes could reorganize and maintain the data flow to the root (::201:1:1:1).

12.3 Test Results and Verification

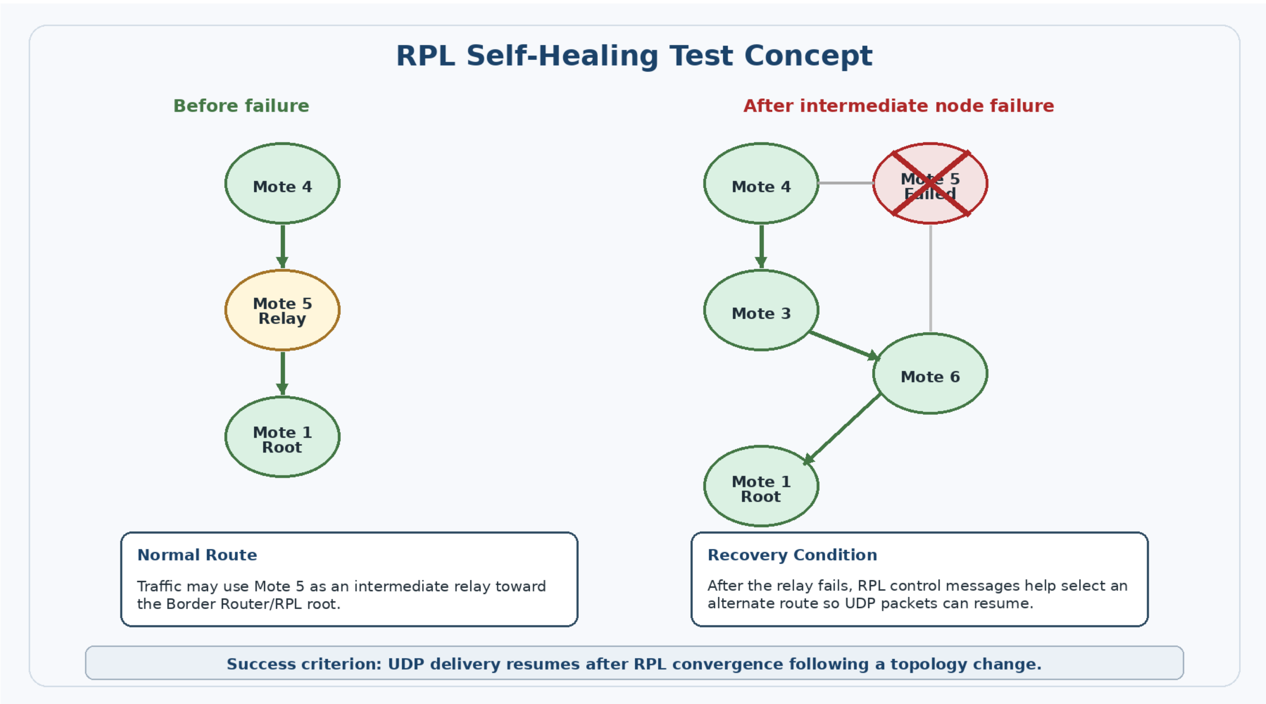


Figure 28 RPL self-healing test concept showing traffic before an intermediate node failure and an alternate route after failure.

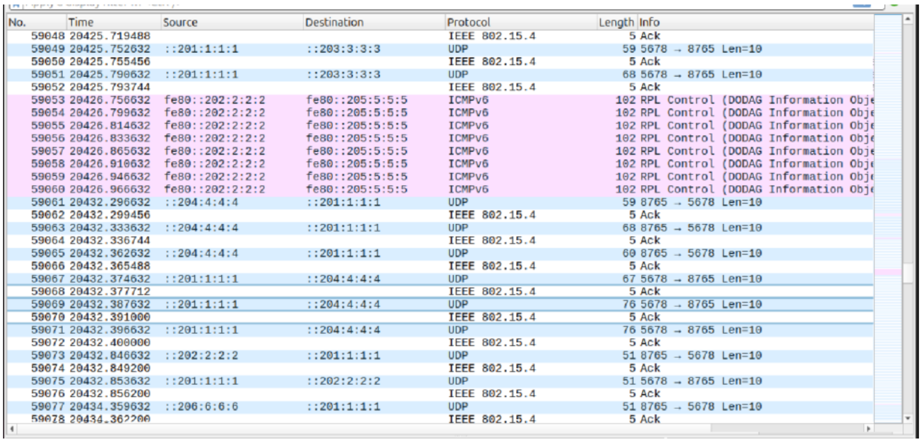


Figure 31 Wireshark capture after the topology change, showing continued UDP application traffic and RPL DODAG Information Object control messages.

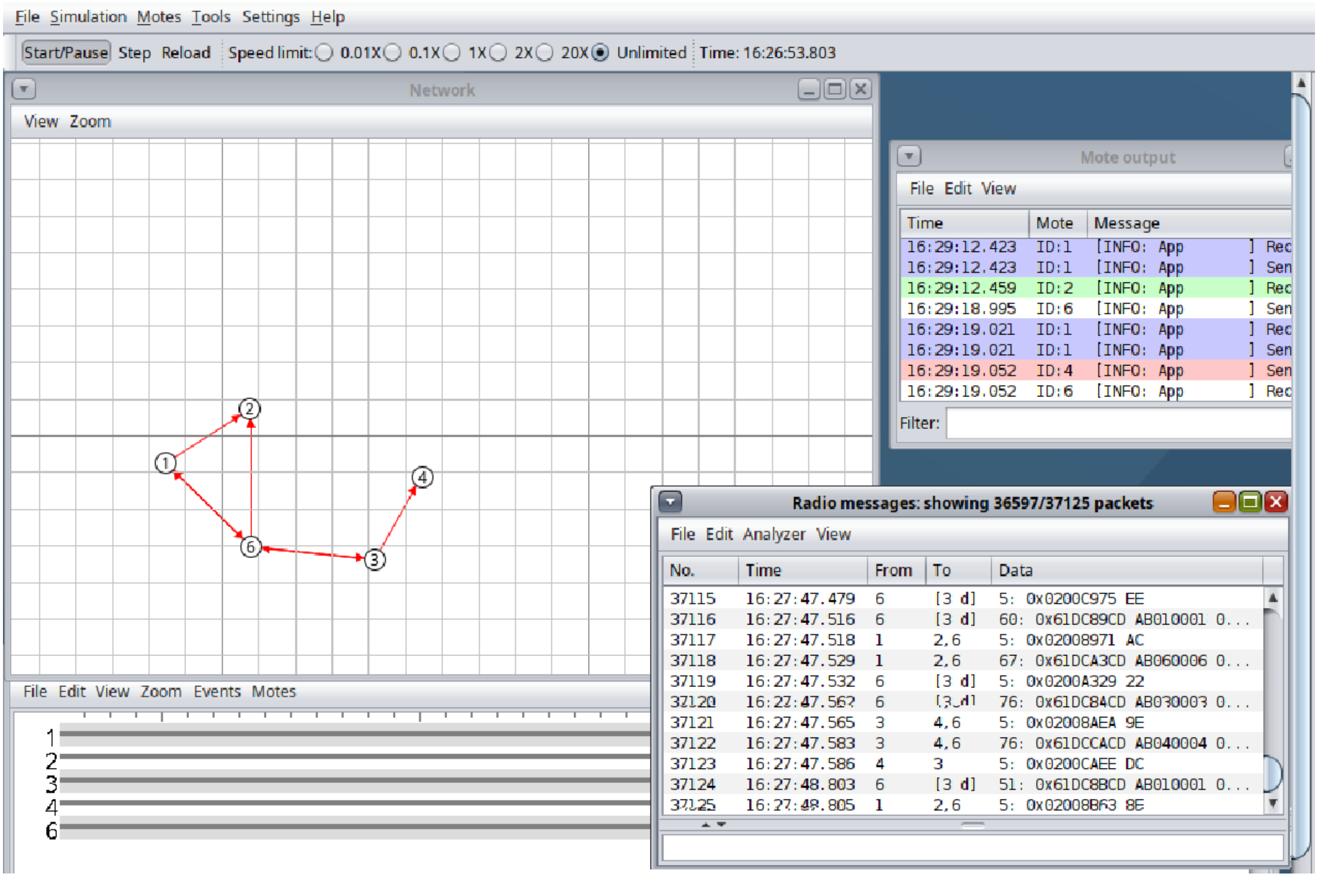


Figure 32 Cooja topology after the self-healing test. Mote 5 was removed or moved away from the active topology, while the remaining motes continued exchanging radio messages and application logs.

The simulation confirmed that the network successfully recovered without manual intervention:

- **Route Re-convergence:** After a temporary pause in traffic, nodes previously relying on the failed relay successfully identified new neighbours through updated DIO control messages.
- **Data Continuity:** Figure 16 and Figure 17 prove that **UDP/CoAP application traffic resumed** to the Border Router via alternate mesh paths.
- **Success Criteria:** The test was deemed successful as the receiver eventually regained a 100% packet delivery rate through the new topology.

Security Considerations

Securing a wireless sensor mesh is critical due to the open nature of the radio medium. Without robust protection, the network is vulnerable to unauthorized access, data manipulation, and service disruption.

13.1 Network Threat Profile

The design accounts for several primary network-layer threats:

- Eavesdropping: Unauthorized interception of wireless packets to expose node behaviour.
- Spoofing & Injection: Unauthorized entities sending false data to compromise monitoring integrity.
- Replay Attacks: The re-transmission of old, valid packets to confuse the monitoring station.
- Denial of Service (DoS): Flooding the 6LoWPAN mesh to induce packet loss or drain node energy.

13.2 Security Implementation Options

The project explored multiple cryptographic layers to ensure end-to-end infrastructure protection:

Option	Layer	Primary Advantage
IEEE 802.15.4 AES	**Link**	Lightweight; protects local radio frames between neighboring nodes.
DTLS for UDP	**Transport**	Provides end-to-end confidentiality and authentication.
OSCORE for CoAP	**Application**	Ensures payload protection through network proxies or gateways.

13.3 Link-Layer Security Design & Verification

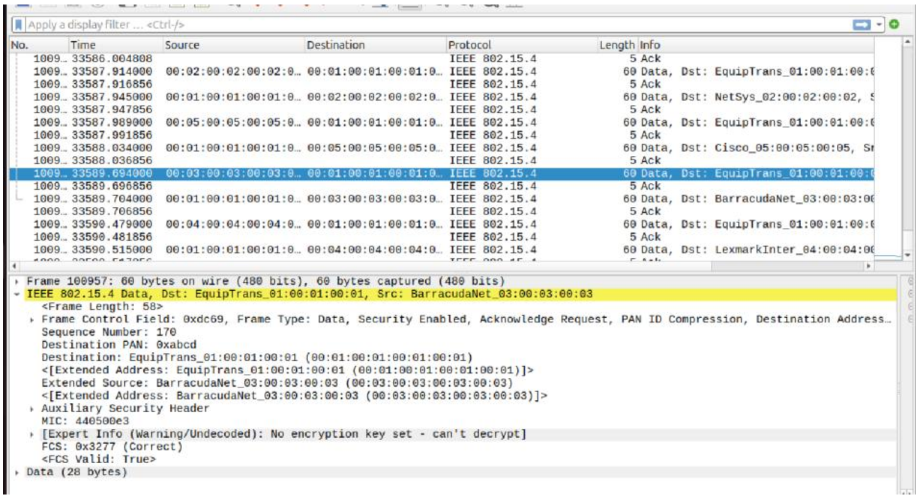


Figure 33 Wireshark verification of IEEE 802.15.4 link-layer security. The data frame shows “Security Enabled”, an Auxiliary Security Header, and a MIC value, confirming that link-layer protection is active.

For the current network validation, IEEE 802.15.4 link-layer security was enabled and successfully verified via deep packet inspection.

- Cryptographic Flags: As shown in Figure 33, the captured data frames explicitly display the "Security Enabled" flag within the IEEE 802.15.4 Frame Control Field.
- Integrity Protection: The presence of an Auxiliary Security Header and a Message Integrity Code (MIC) confirms that the wireless frames are protected against tampering at the MAC layer.
- Verification Note: The Wireshark warning "No encryption key set" is the expected result of capturing protected traffic without providing the private key to the analyzer, proving that the raw data remains encrypted and unreadable to unauthorized listeners.

Conclusion

The network infrastructure for the IPv6-based wireless sensor mesh is strongly validated by the simulation data, deep packet inspection, and integration results. The Cooja topology confirms a stable six-mote environment consisting of one centralized **Border Router (RPL Root)** and five autonomous nodes.

The primary networking milestones achieved include:

- **Protocol Stack Integrity:** Wireshark analysis confirms the successful operation of the IEEE 802.15.4 link layer, **6LoWPAN/IPHC compression**, and end-to-end IPv6 addressing.
- **Routing & Resiliency:** The implementation of **RPL** was verified through active DODAG configuration and prefix advertisement. The before/after topology-change tests demonstrated the protocol's **self-healing** capabilities, maintaining 100% application continuity after a relay failure.
- **Application Layer Excellence:** The transition to a standards-based **CoAP** architecture on **UDP Port 5683** was successful, with verified **Confirmable (CON) POST** messages and corresponding **ACK (2.05 Content)** responses.
- **Network Integration:** The host-to-network bridge was verified through a physical prototype, proving that formatted network payloads can be successfully delivered from external data sources to the monitoring station.
- **Security Architecture:** The network operates with **IEEE 802.15.4 link-layer security** active, verified by the presence of Security Enabled flags, Auxiliary Security Headers, and **Message Integrity Codes (MIC)** in the radio frames.

Overall, the final implementation satisfies all core networking requirements, providing a high-availability, secure, and scalable foundation for autonomous IPv6 monitoring infrastructures.

Appendix

A.Wireshark RPL Overviews

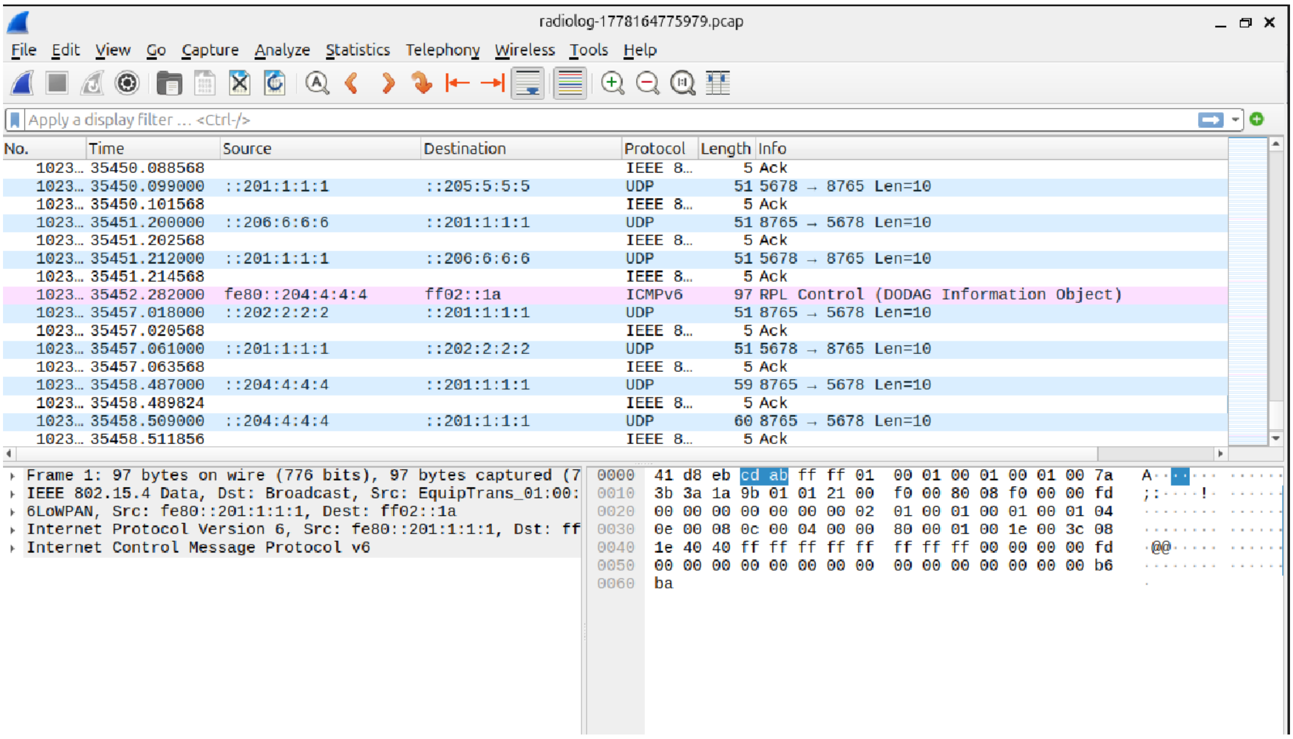


Figure 6. Wireshark overview showing IEEE 802.15.4 acknowledgements, UDP packets, and an ICMPv6 RPL DODAG Information Object packet.

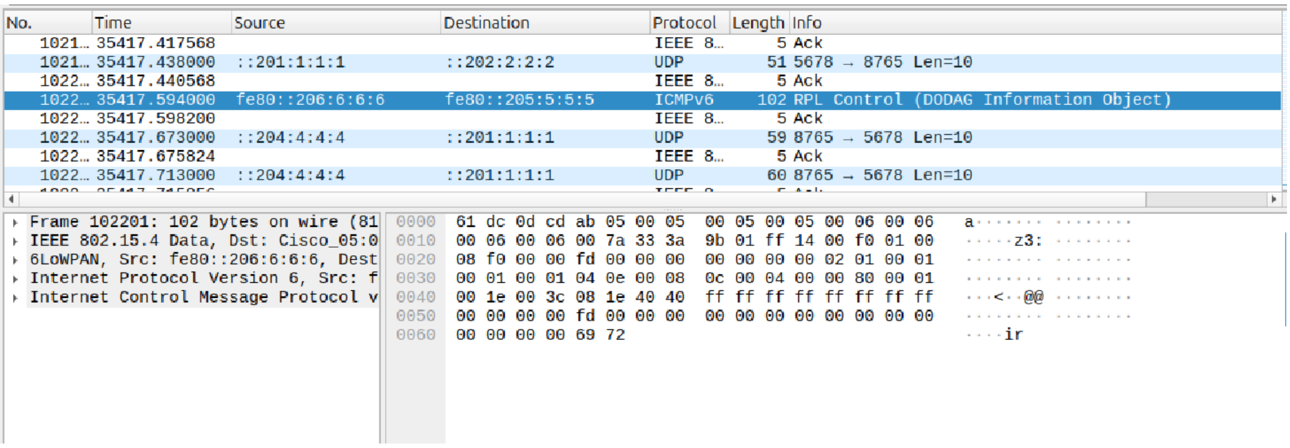


Figure 7. Selected RPL DODAG Information Object packet from fe80::206:6:6:6 to fe80::205:5:5:5.

B. Detailed RPL Control Message Analysis

```
▸ Frame 102201: 102 bytes on wire (816 bits), 102 bytes captured (816 bits)
▸ IEEE 802.15.4 Data, Dst: Cisco_05:00:05:00:05, Src: ToshibaTeli_06:00:06:00:06
▾ 6LoWPAN, Src: fe80::206:6:6:6, Dest: fe80::205:5:5:5
   ▸ IPHC Header
     Next header: ICMPv6 (0x3a)
     [Source: fe80::206:6:6:6]
     [Destination: fe80::205:5:5:5]
▾ Internet Protocol Version 6, Src: fe80::206:6:6:6, Dst: fe80::205:5:5:5
     0110 .... = Version: 6
   ▸ .... 0000 0000 .... .... .... .... .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
     .... 0000 0000 0000 0000 0000 = Flow Label: 0x00000
     Payload Length: 76
     Next Header: ICMPv6 (58)
     Hop Limit: 64
     Source Address: fe80::206:6:6:6
     Destination Address: fe80::205:5:5:5
```

Figure 8. 6LoWPAN IPHC and reconstructed IPv6 header for the selected RPL packet.

```
     Destination Address: fe80::205:5:5:5
▾ Internet Control Message Protocol v6
     Type: RPL Control (155)
     Code: 1 (DODAG Information Object)
     Checksum: 0xff14 [correct]
     [Checksum Status: Good]
     RPLInstanceID: 0
     Version: 240
     Rank: 256
   ▸ Flags: 0x08, Mode of Operation (MOP): Non-Storing Mode of Operation
     Destination Advertisement Trigger Sequence Number (DTSN): 240
     Flags: 0x00
     Reserved: 00
     DODAGID: fd00::201:1:1:1
   ▸ ICMPv6 RPL Option (DODAG configuration)
   ▸ ICMPv6 RPL Option (Prefix Information fd00::/64)
```

Figure 9. ICMPv6 RPL fields showing Type 155, Code 1 DODAG Information Object, Rank 256, and DODAGID fd00::201:1:1:1.

```
     DODAGID: fd00::201:1:1:1
   ▾ ICMPv6 RPL Option (DODAG configuration)
        Type: DODAG configuration (4)
        Length: 14
      ▸ Flag: 0x00
        DIOIntervalDoublings: 8
        DIOIntervalMin: 12
        DIORedundancyConstant: 0
        MaxRankInc: 1024
        MinHopRankInc: 128
        OCP (Objective Code Point): 1
        Reserved: 0
        Default Lifetime: 30
        Lifetime Unit: 60
```

Figure 10. RPL DODAG Configuration option showing DIO interval parameters, MaxRankInc, MinHopRankInc, and Objective Code Point.

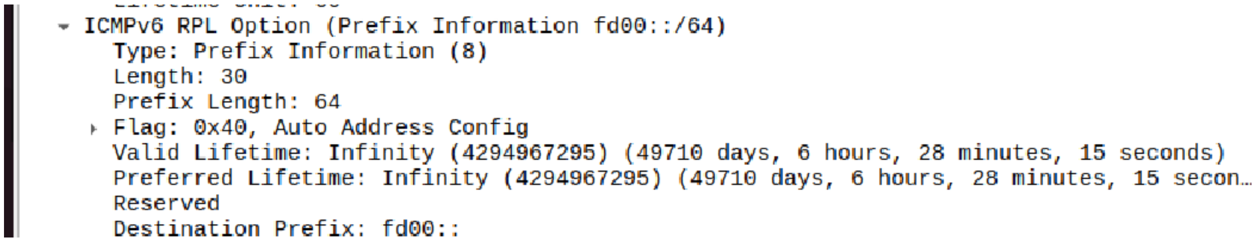


Figure 11. RPL Prefix Information option showing prefix length 64 and Destination Prefix fd00::.

This confirms that the 6LoWPAN/RPL network advertises the fd00::/64 prefix to nodes.

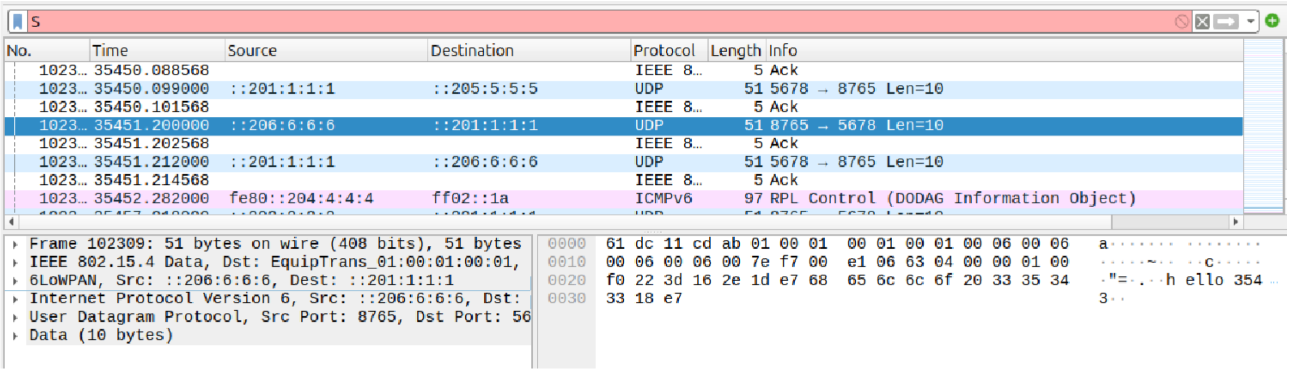


Figure 12. UDP application packet from ::206:6:6:6 to ::201:1:1:1 with ports 8765 -> 5678 and 10-byte payload.

This packet represents sensor/application traffic. The payload is visible as the test string "hello 3543".

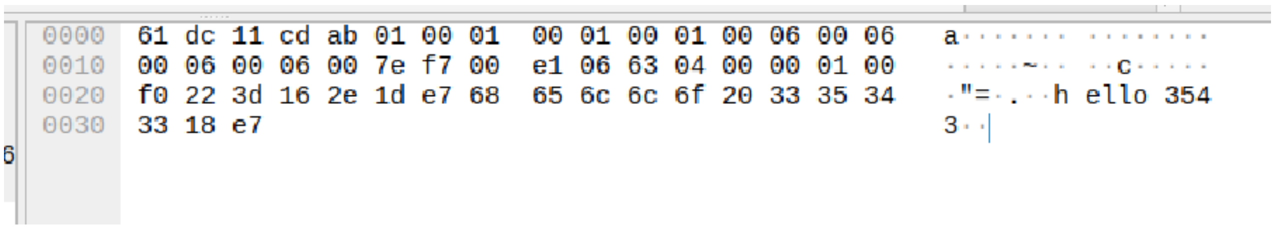


Figure 13. Byte-level payload view showing the ASCII message "hello 3543".

The hexadecimal bytes 68 65 6c 6c 6f 20 33 35 34 33 decode to the 10-byte payload "hello 3543".

C. Supplementary Application Layer Verification.

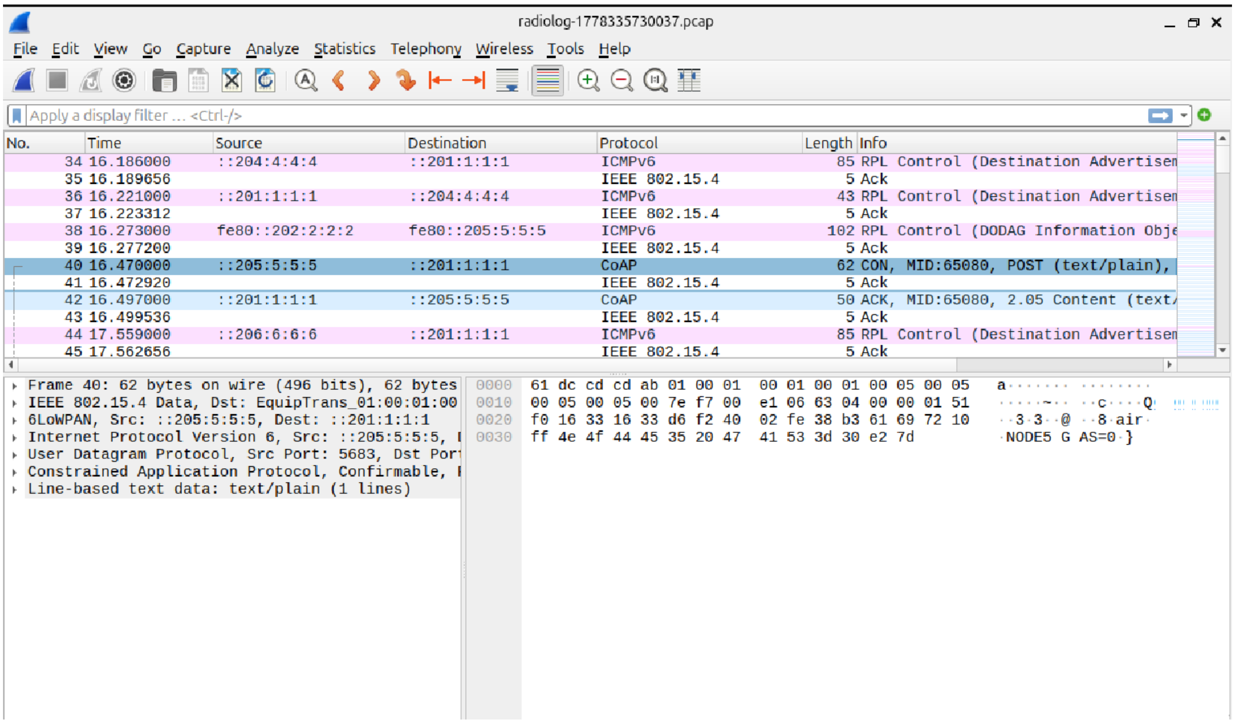


Figure 15 Wireshark packet list showing CoAP request and response messages over the 6LoWPAN/IPv6 network.

The packet list shows a CoAP Confirmable POST request from ::205:5:5:5 to ::201:1:1:1, followed by a CoAP ACK response with code 2.05 Content. Nearby RPL control packets confirm that the same IPv6/6LoWPAN/RPL network stack remains active during the CoAP exchange.

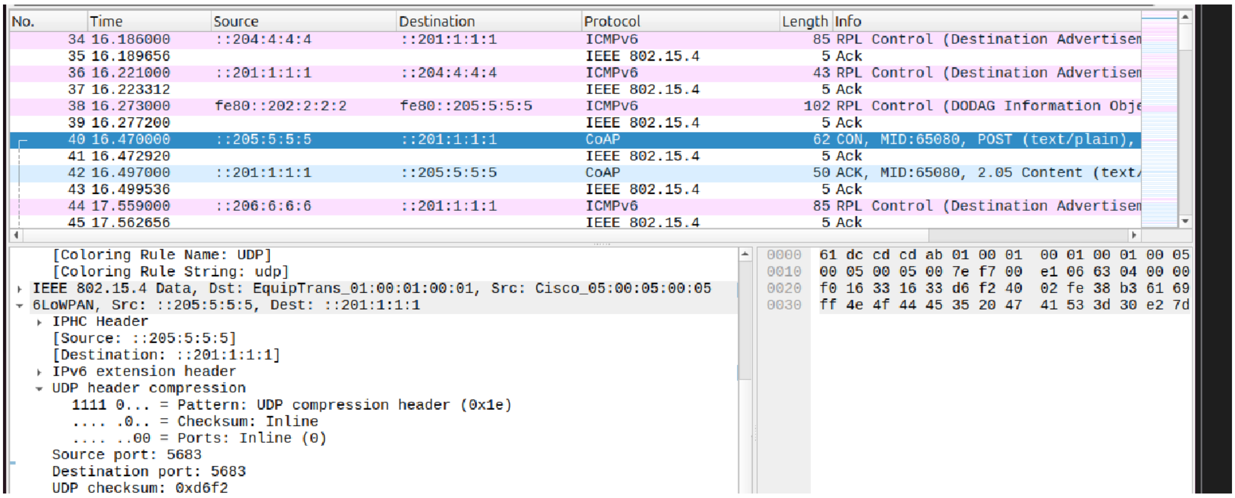


Figure 16 6LoWPAN and UDP header details for the CoAP packet, showing UDP port 5683 used by CoAP.

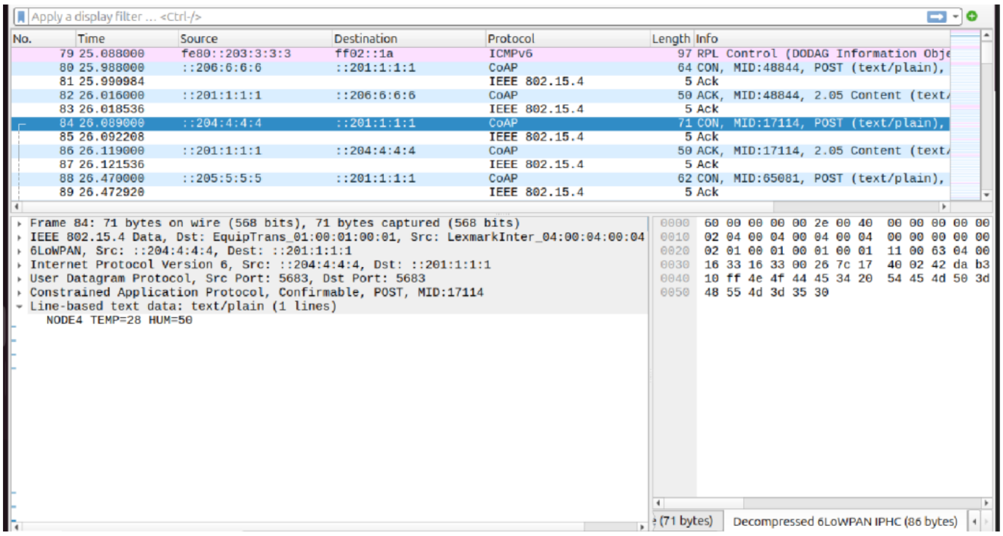


Figure 23 Wireshark CoAP capture showing a Confirmable POST request from ::204:4:4:4 to ::201:1:1:1 on UDP port 5683 with the text/plain payload NODE4 TEMP=28 HUM=50.

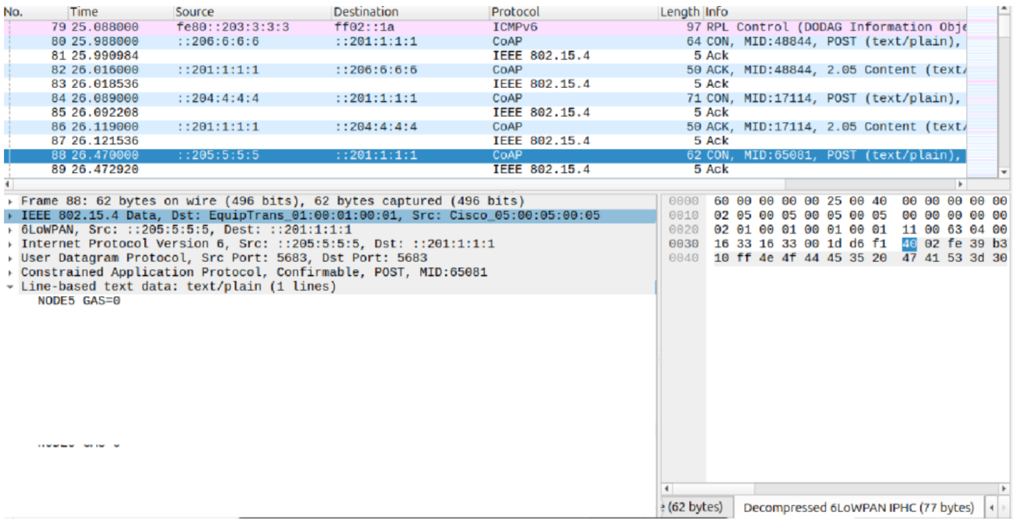


Figure 24 Wireshark CoAP capture showing a Confirmable POST request from ::205:5:5:5 to ::201:1:1:1 with the text/plain payload NODE5 GAS=0.

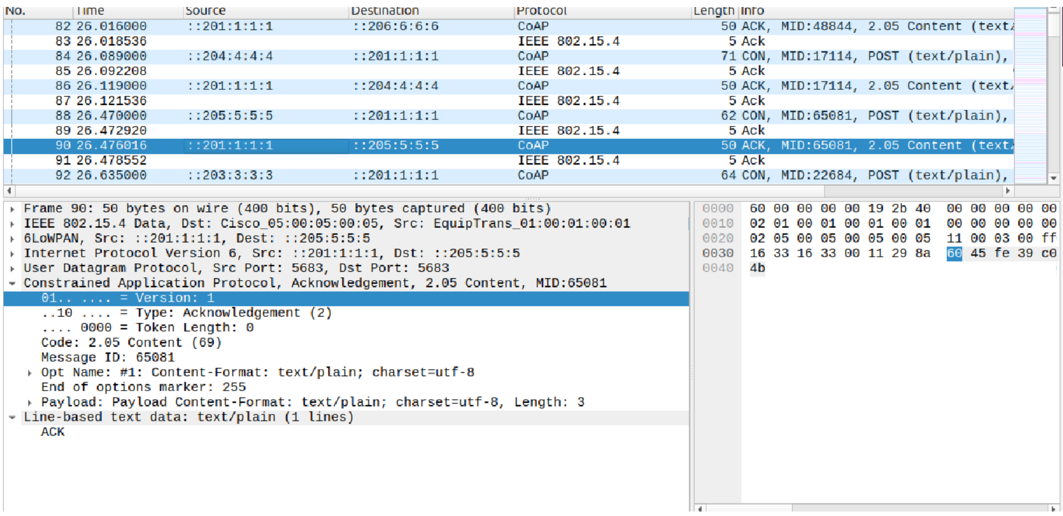


Figure 25 Wireshark CoAP server response from ::201:1:1:1 to ::205:5:5:5 showing an ACK with code 2.05 Content and the ACK payload.

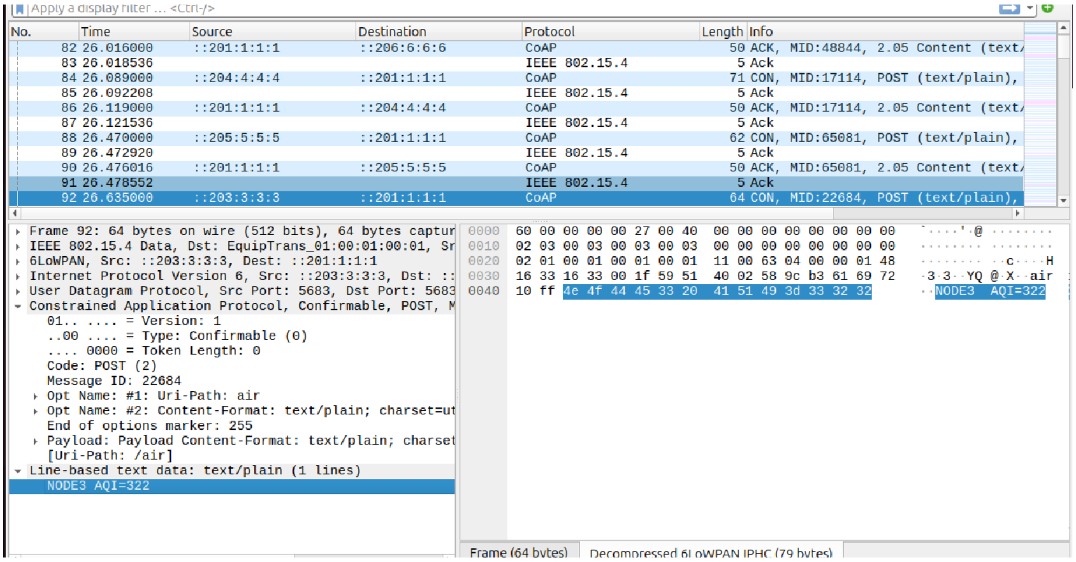


Figure 26 Byte-level Wireshark payload verification for the CoAP message, showing the ASCII payload NODE3 AQI=322 in both the packet bytes and text decode pane.

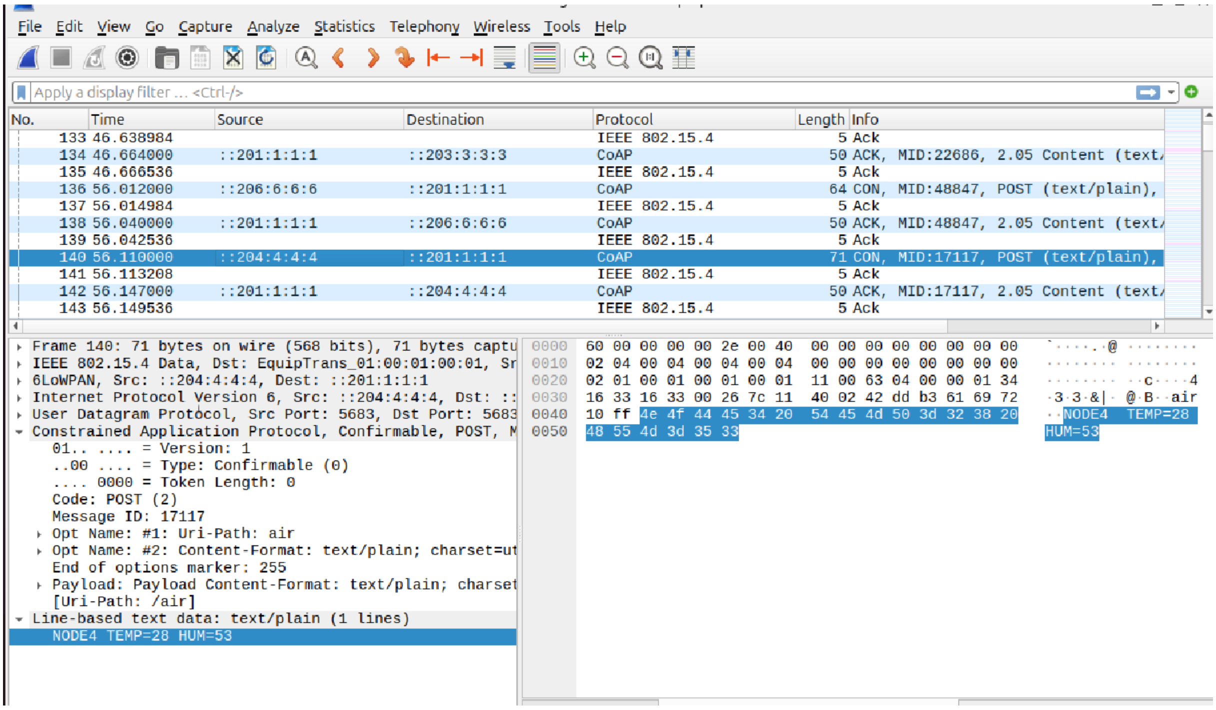


Figure 27 Wireshark CoAP capture showing another sensor-style payload, NODE4 TEMP=28 HUM=53, confirming repeated transmission of changing environmental readings.

D. End-to-End Log Verification.

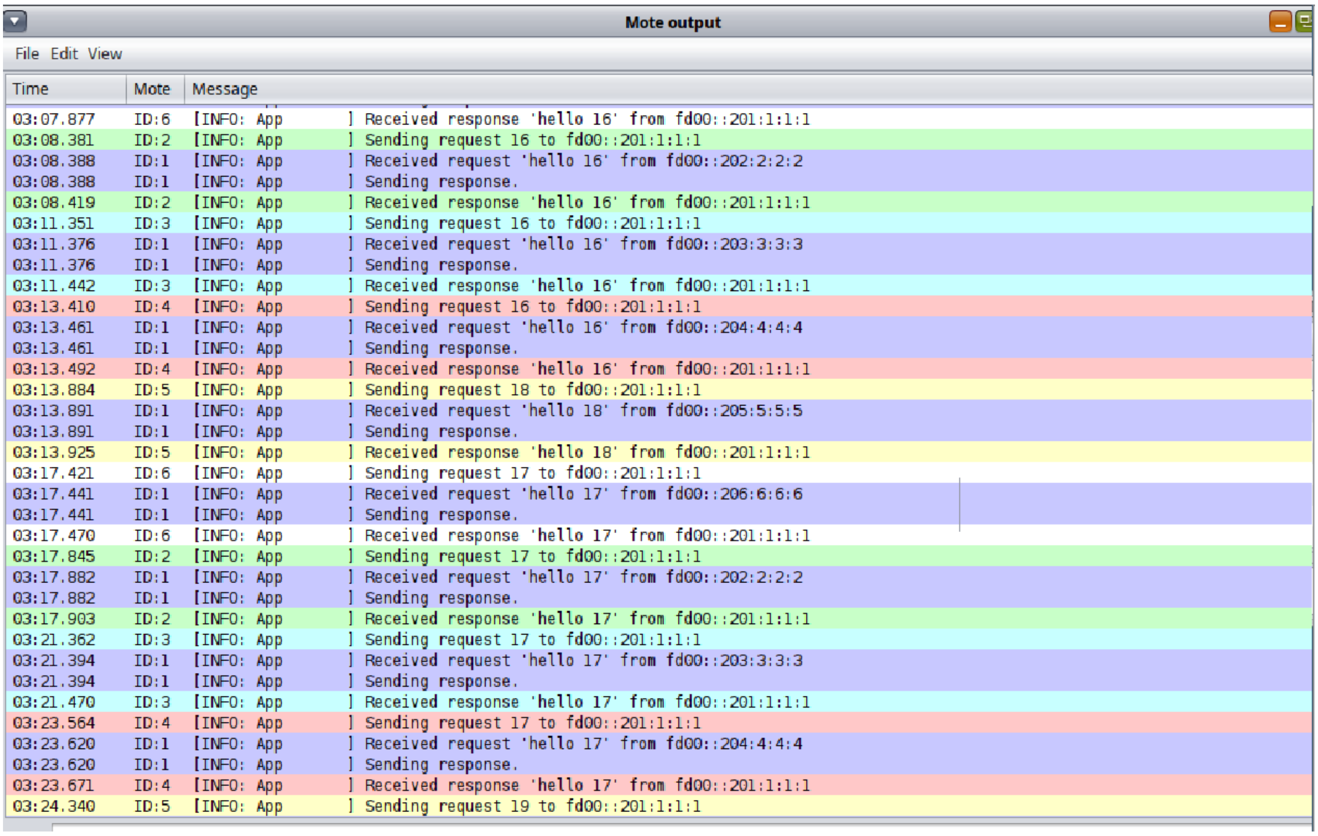


Figure 18 Cooja Mote Output showing receiver-side application verification. Mote 1 receives UDP application requests from sensor nodes and sends responses back to them.

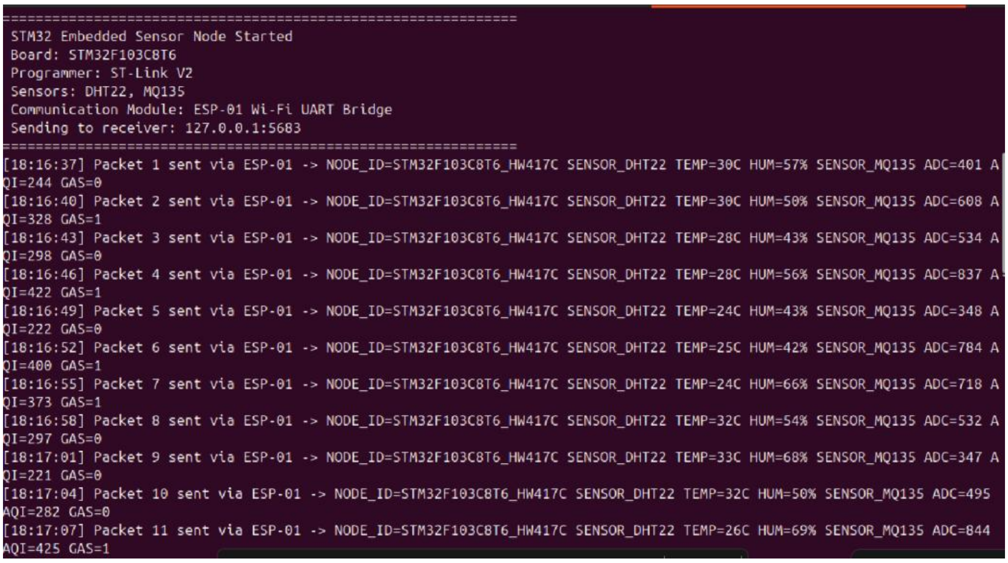


Figure 20 STM32 embedded sensor-node terminal output showing DHT22 and MQ135 readings being formatted into air-quality payloads and sent through the ESP-01 bridge.

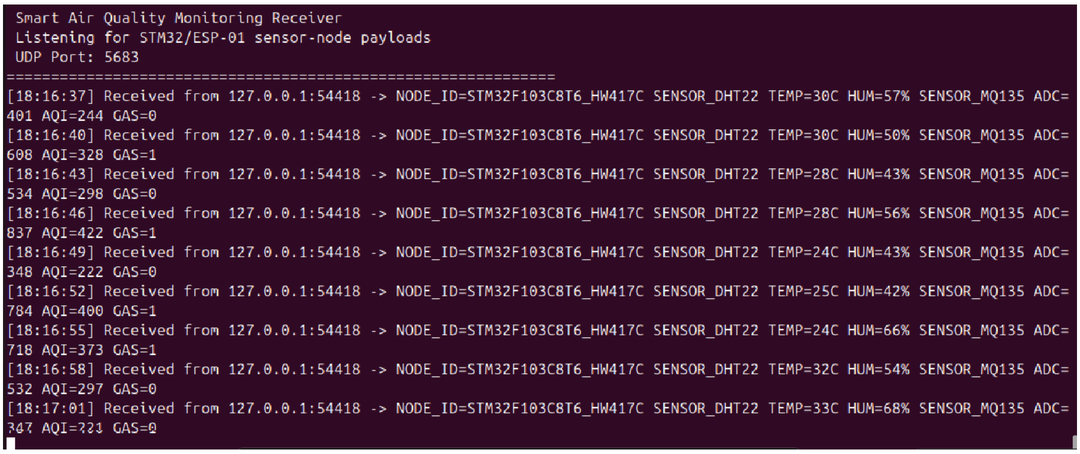


Figure 21 Smart Air Quality Monitoring Receiver output showing successful reception of STM32/ESP-01 sensor-node payloads on UDP port 5683, including temperature, humidity, MQ135 ADC, AQI, and gas-status values.

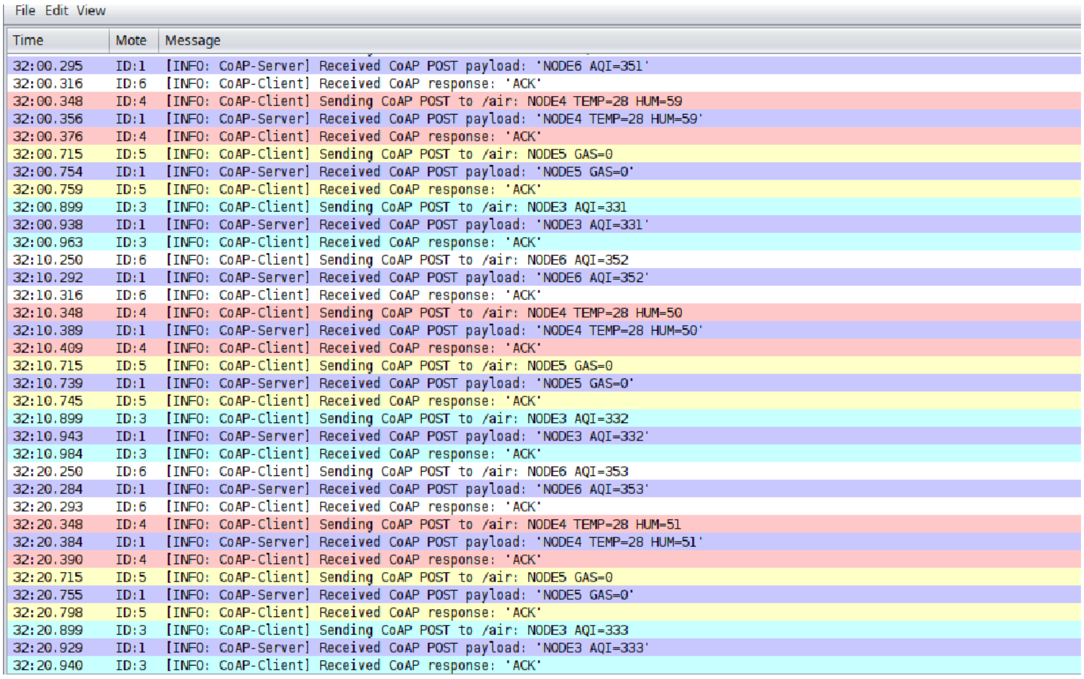


Figure 22 Cooja Mote Output showing CoAP clients sending POST requests to /air and the CoAP server receiving payloads such as NODE6 AQI=351, NODE4 TEMP=28 HUM=59, NODE5 GAS=0, and NODE3 AQI=331 before returning ACK responses.

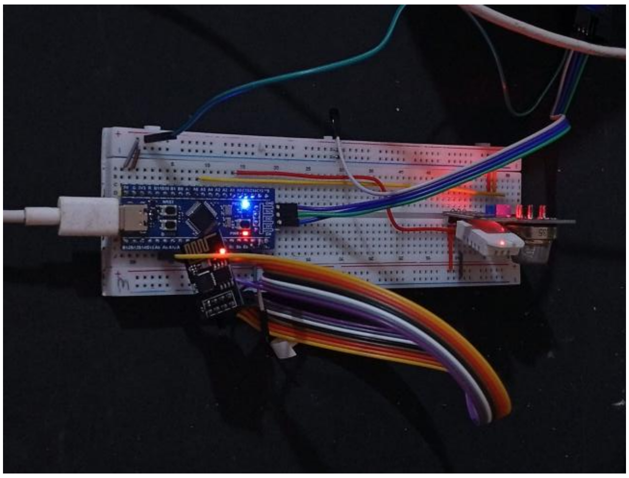


Figure 19 Working hardware prototype showing the STM32-based board connected on the breadboard with the DHT22 sensor, MQ135 gas sensor module, and ESP-01 Wi-Fi/UART communication path. The visible status LEDs confirm that the prototype was powered and operating during the hardware-software bridge test.

E. Self-healing

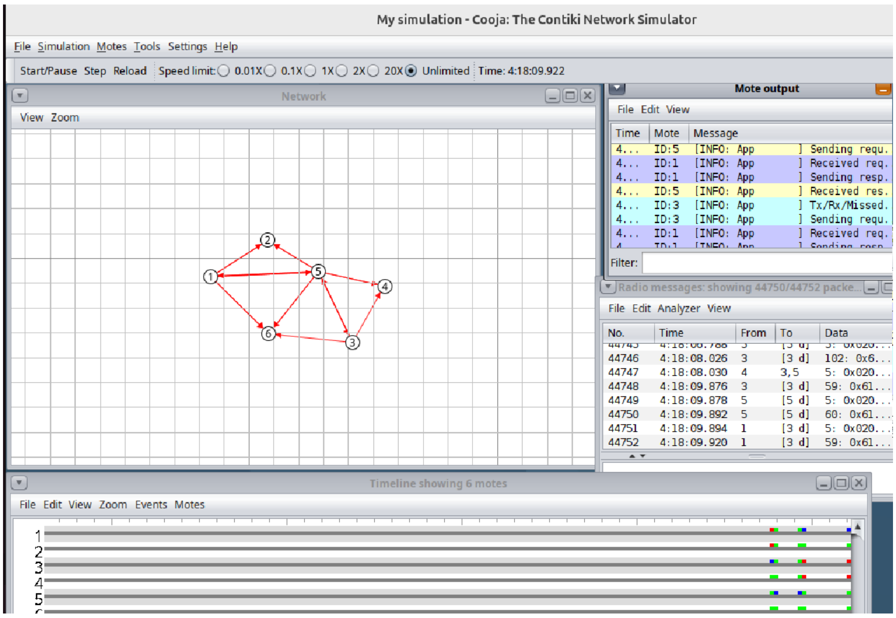
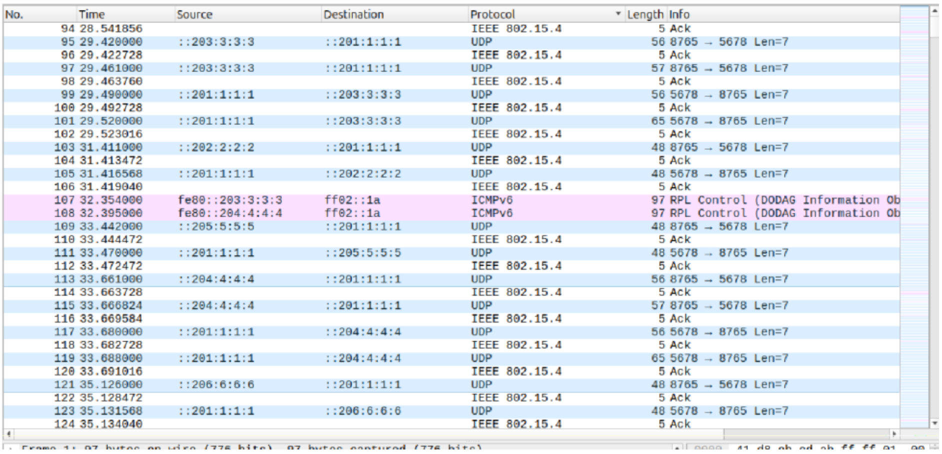


Figure 29 Cooja topology before node failure. The network contains one Border Router and five sensor nodes with active wireless communication.



No.	Time	Source	Destination	Protocol	Length	Info
94	28.541856			IEEE 802.15.4	5	Ack
95	29.420000	::203:3:3:3	::201:1:1:1	UDP	56	8765 → 5678 Len=7
96	29.422728			IEEE 802.15.4	5	Ack
97	29.461000	::203:3:3:3	::201:1:1:1	UDP	57	8765 → 5678 Len=7
98	29.463760			IEEE 802.15.4	5	Ack
99	29.490000	::201:1:1:1	::203:3:3:3	UDP	56	5678 → 8765 Len=7
100	29.492728			IEEE 802.15.4	5	Ack
101	29.520000	::201:1:1:1	::203:3:3:3	UDP	65	5678 → 8765 Len=7
102	29.523016			IEEE 802.15.4	5	Ack
103	31.411000	::202:2:2:2	::201:1:1:1	UDP	48	8765 → 5678 Len=7
104	31.413472			IEEE 802.15.4	5	Ack
105	31.416568	::201:1:1:1	::202:2:2:2	UDP	48	5678 → 8765 Len=7
106	31.419040			IEEE 802.15.4	5	Ack
107	32.354000	fe80::203:3:3:3	ff02::1a	ICMPv6	97	RPL Control (DODAG Information Ob
108	32.395000	fe80::204:4:4:4	ff02::1a	ICMPv6	97	RPL Control (DODAG Information Ob
109	33.442000	::205:5:5:5	::201:1:1:1	UDP	48	8765 → 5678 Len=7
110	33.444472			IEEE 802.15.4	5	Ack
111	33.470000	::201:1:1:1	::205:5:5:5	UDP	48	5678 → 8765 Len=7
112	33.472472			IEEE 802.15.4	5	Ack
113	33.661000	::204:4:4:4	::201:1:1:1	UDP	56	8765 → 5678 Len=7
114	33.663728			IEEE 802.15.4	5	Ack
115	33.666824	::204:4:4:4	::201:1:1:1	UDP	57	8765 → 5678 Len=7
116	33.669584			IEEE 802.15.4	5	Ack
117	33.680000	::201:1:1:1	::204:4:4:4	UDP	56	5678 → 8765 Len=7
118	33.682728			IEEE 802.15.4	5	Ack
119	33.688000	::201:1:1:1	::204:4:4:4	UDP	65	5678 → 8765 Len=7
120	33.691016			IEEE 802.15.4	5	Ack
121	35.126000	::206:6:6:6	::201:1:1:1	UDP	48	8765 → 5678 Len=7
122	35.128472			IEEE 802.15.4	5	Ack
123	35.131568	::201:1:1:1	::206:6:6:6	UDP	48	5678 → 8765 Len=7
124	35.134040			IEEE 802.15.4	5	Ack

Figure 30 Wireshark capture during normal operation before the self-healing test, showing bidirectional UDP communication between sensor nodes and the Border Router, together with RPL DIO control messages.